7CS039 Lab 1 Descriptive Statistics

1 Objectives

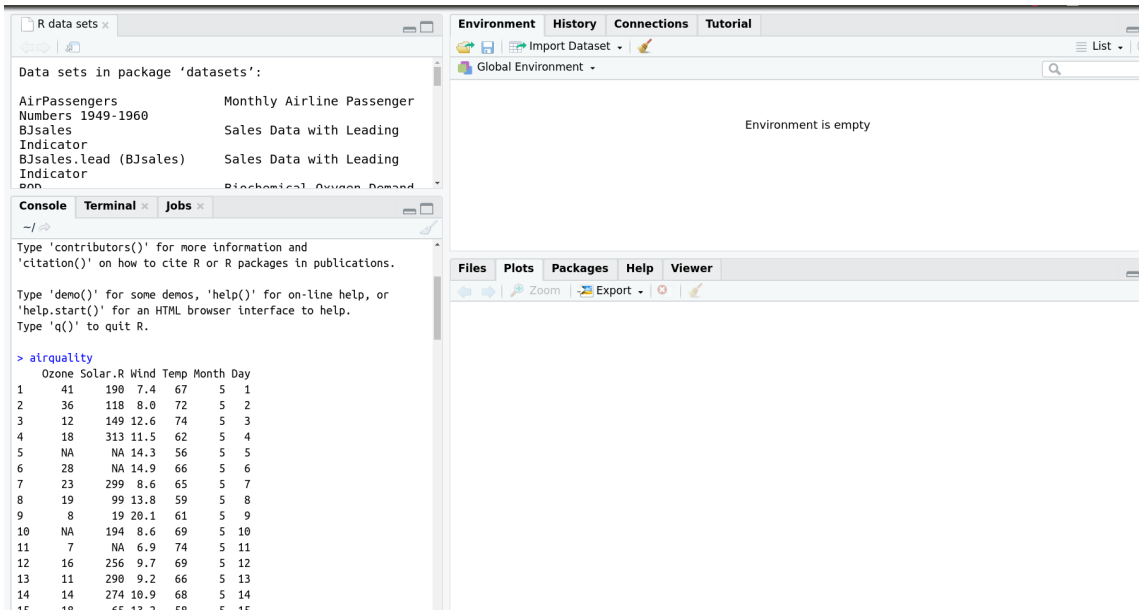
By the end of this session we will be able to

- Create summary statistics for a single group and by different groups
- Generate graphical display of data such as histograms, box plots, bar plots, dot charts and pie charts etc.

2 Let's get started

We'll be using the `airquality` data set for this par of the session. It's one of the built in data sets. To view it type

```
> airquality
```



This data set records daily air quality measurements in New York from May to September 1973. Daily readings of the following air quality values for May 1, 1973 to September 30, 1973 are provided

- Ozone: Mean ozone in parts per billion from 1300 to 1500 hours at Roosevelt Island
- Solar.R: Solar radiation in Langleys in the frequency band 4000–7700 Angstroms from 0800 to 1200 hours at Central Park
- Wind: Average wind speed in miles per hour at 0700 and 1000 hours at LaGuardia Airport
- Temp: Maximum daily temperature in degrees Fahrenheit at La Guardia Airport.

There are lots of really nice datasets included in R for you to play around with. Most have been chosen because they exhibit nice properties. To view the first few (5 below) lines (rather than all of the entries) we do

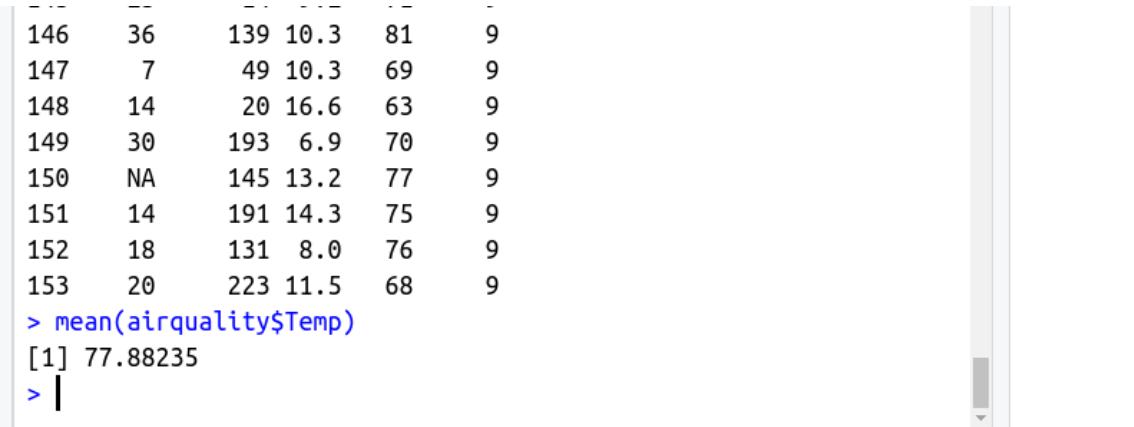
```
> airquality[1:5,]
```

3 Summary Statistics

3.1 The Mean

To compute the mean temperature reading we do

```
> mean(airquality$Temp)
```



Exercise

Find the mean reading for the variables `Ozone` and `Solar.R`.

3.1.1 Not Available Values

There are some days where particular readings are not available (marked by `NA`) in the data. To compute descriptive statistics by first removing these values we use the optional argument `na.rm=T`

```
> mean(airquality$Temp, na.rm=T)
```

Exercise

Find the mean reading for all of the appropriate variables by firstly removing the `NA` values.

3.2 Variance & Standard Deviation

To compute the variance and standard deviation of the mean temperature reading we do

```
> var(airquality$Temp)
```

```
> var(airquality$Temp)
[1] 89.59133
>
```

```
> sd(airquality$Temp)
```

```
> var(airquality$Temp)
[1] 89.59133
> sd(airquality$Temp)
[1] 9.46527
>
```

3.3 The Median

To compute the median temperature reading we do

```
> median(airquality$Temp)
```

```
> mean(airquality$Ozone)
[1] NA
> mean(airquality$Wind)
[1] 9.957516
> median(airquality$Temp)
[1] 79
> airquality[1:5,]
  Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
> median(airquality$Temp)
[1] 79
>
```

3.4 Five Number Summary

To compute the five number summary for the variable `Temp`

```
> summary(airquality$Temp)
```

```
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
> median(airquality$Temp)
[1] 79
> summary(airquality$Temp)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  56.00   72.00   79.00   77.88   85.00   97.00
>
```

The `summary` function automatically removes `NA` values. We can produce a five number summary for all of the variables In a single attempt by doing

```
> summary(airquality)
```

```
> summary(airquality)
     Ozone           Solar.R           Wind
 Min.   :  1.00   Min.   :  7.0   Min.   : 1.700
 1st Qu.: 18.00   1st Qu.:115.8   1st Qu.: 7.400
 Median : 31.50   Median :205.0   Median : 9.700
 Mean   : 42.13   Mean   :185.9   Mean   : 9.958
 3rd Qu.: 63.25   3rd Qu.:258.8   3rd Qu.:11.500
 Max.   :168.00   Max.   :334.0   Max.   :20.700
 NA's   :37       NA's   :7
      Temp           Month            Day
 Min.   :56.00   Min.   :5.000   Min.   : 1.0
 1st Qu.:72.00   1st Qu.:6.000   1st Qu.: 8.0
 Median :79.00   Median :7.000   Median :16.0
 Mean   :77.88   Mean   :6.993   Mean   :15.8
 3rd Qu.:85.00   3rd Qu.:8.000   3rd Qu.:23.0
 Max.   :97.00   Max.   :9.000   Max.   :31.0

>
```

Notice that "Mont" and"Day" are coded as numeric variables even though they are clearly nominal/categorical. This can be mended as follows

```
> airquality$Month = factor(airquality$month)
> summary(airquality)
```

```
> airquality$Month = factor(airquality$Month)
> summary(airquality)
     Ozone           Solar.R           Wind       
 Min.   :  1.00   Min.   :  7.0   Min.   : 1.700  
 1st Qu.: 18.00   1st Qu.:115.8   1st Qu.: 7.400  
 Median : 31.50   Median :205.0   Median : 9.700  
 Mean   : 42.13   Mean   :185.9   Mean   : 9.958  
 3rd Qu.: 63.25   3rd Qu.:258.8   3rd Qu.:11.500  
 Max.   :168.00   Max.   :334.0   Max.   :20.700  
 NA's   :37       NA's   :7                       
      Temp       Month       Day      
 Min.   :56.00   5:31   Min.   : 1.0  
 1st Qu.:72.00   6:30   1st Qu.: 8.0  
 Median :79.00   7:31   Median :16.0  
 Mean   :77.88   8:31   Mean   :15.8  
 3rd Qu.:85.00   9:30   3rd Qu.:23.0  
 Max.   :97.00          Max.   :31.0  

> |
```

Can you explain what the output means now?

Exercise

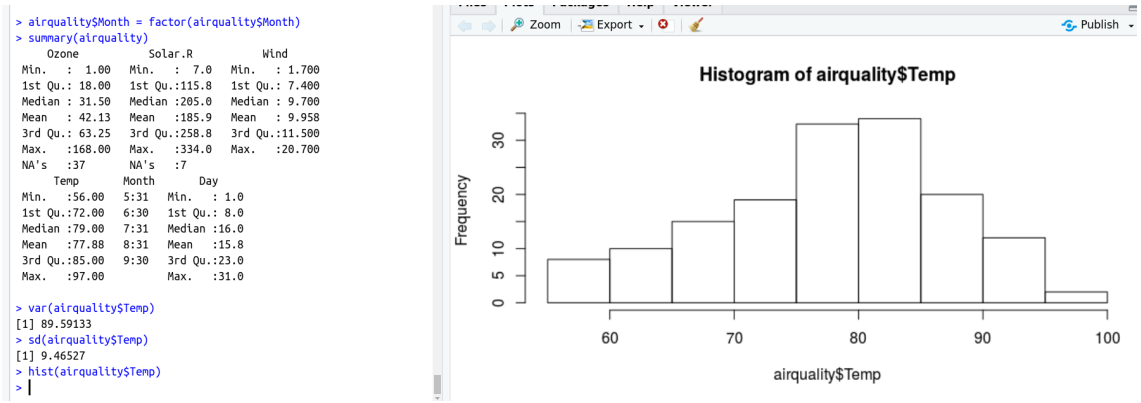
Modify the output appropriately for any nominal/categorical variables

4 Graphical Displays

4.1 Histogram

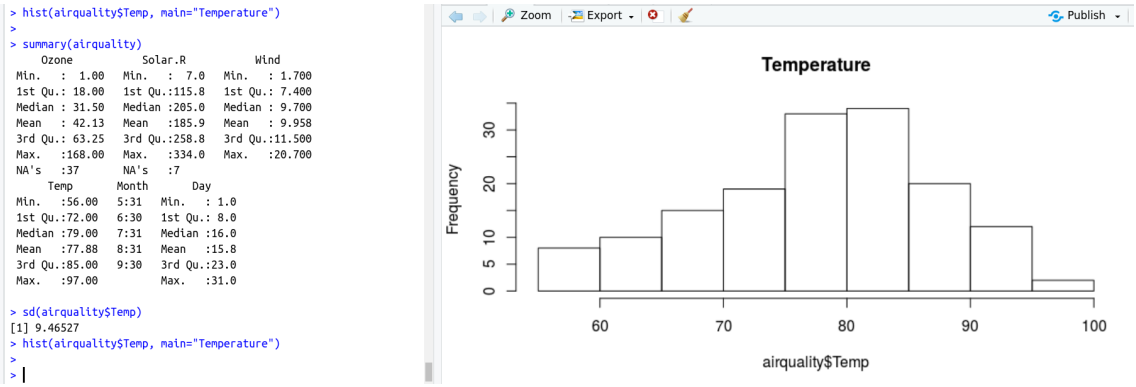
The simplest display for the shape of a distribution of data can be done using a histogram

```
> hist(airquality$Temp)
```



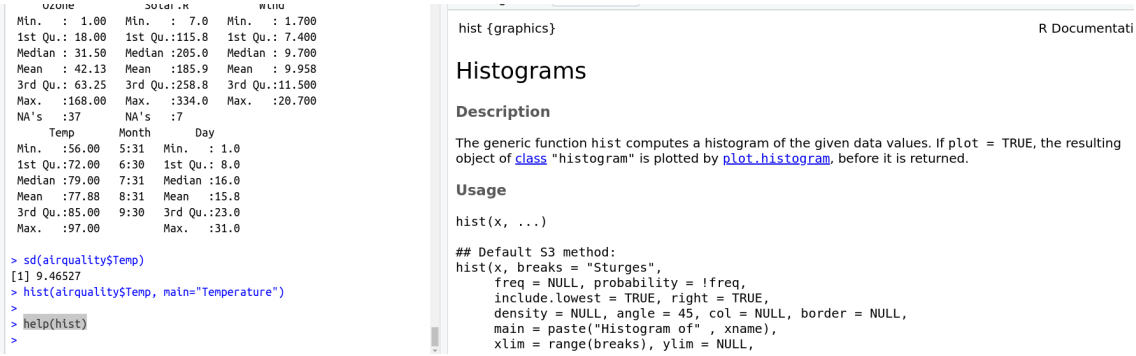
There are lots of options to jazz up your histogram. You can use the internet to help you to do this. For example, to add a title we do

```
> hist(airquality$Temp, main="Temperature")
```



To see an extensive list of the options available for a histogram you can do

```
> help(hist)
```

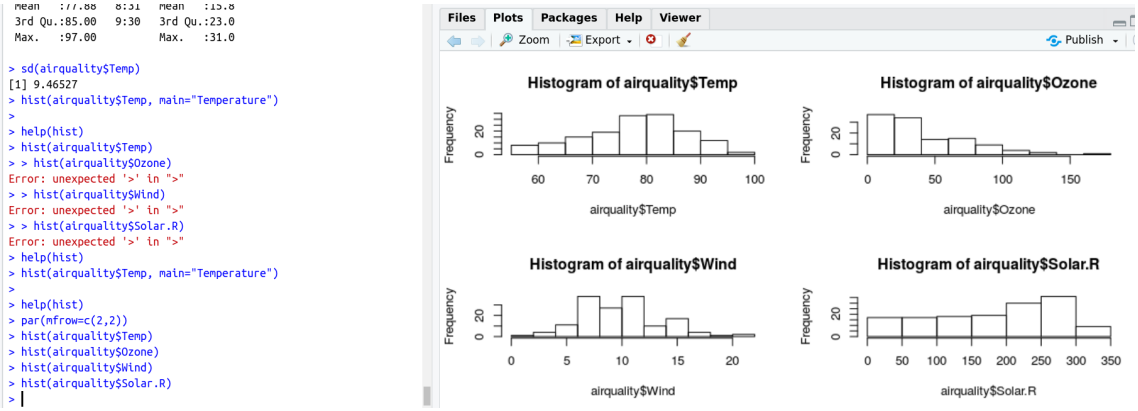


If you want two or more plots in the same window, you can use the command

```
> par(mfrow=c(2,2))
```

which displays the info in two rows and two columns.

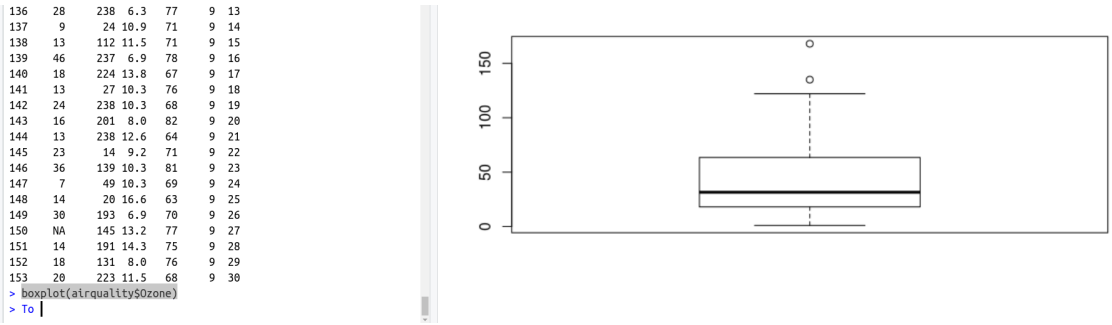
```
> hist(airquality$Temp)
> hist(airquality$Ozone)
> hist(airquality$Wind)
> hist(airquality$Solar.R)
```



4.2 Box Plots

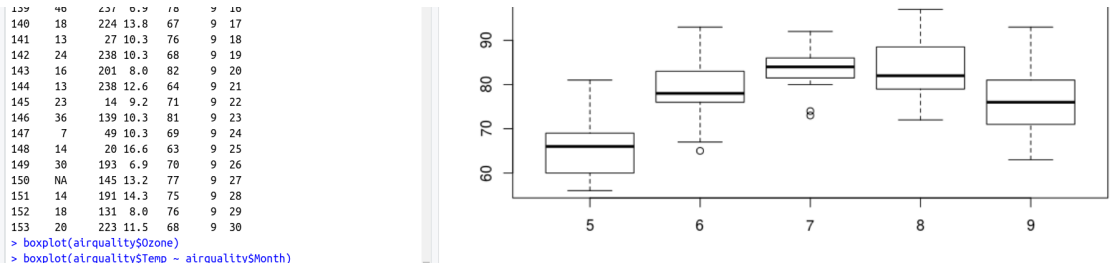
These are one of my favourites. To create a boxplot we do

```
> boxplot(airquality$Ozone)
```



The real power of box plots is to do comparisons of variables by grouping. For example, we may be interested in comparing the fluctuations in temperature across months. To create boxplots of temperature data grouped by the factor "month", we do

```
> boxplot(airquality$Temp ~ airquality$Month)
```



There are a bunch of options for boxplots, just like with histograms, so that you can make them as jazzy as you like. Use the internet to help you.

5 Exercise

Exercises

Choose one of the other built in data sets and produce the descriptive statistics and graphical displays above. Comment on any interesting features of the charts you produce.

We'll spend twenty minutes on this task, then we'll do something else fun!

6 Using R for Data Science -Some fun with GGPlot

R has several systems for making graphs, but `ggplot2` is one of the most elegant and most versatile. `ggplot2` implements the grammar of graphics, a coherent system for describing and building graphs. With `ggplot2`, you can do more faster by learning one system and applying it in many places. This section of the workshop will the R for Data Science page below

https://r4ds.had.co.nz/data-visualisation.html

As MSc students it's important that you develop the skills to go beyond what we learn in the classroom to increase and enhance your own knowledge. That's why we'll finish each workshop by using an externals source.